Data Corrections for Lower Limit H-α Luminosity Functions at 2<z<7 with Pandeia

Rahul Shaji
University of Texas at Austin

Abstract

Observations of star formation rate density (SFRD), the total mass of stars formed per year per unit volume, are key for constraining galaxy evolution over early cosmic time. Galaxies that emit H-α spectral emission lines are strong tracers of star formation. Thus, calculating the number densities of H-α detected galaxies at different redshifts as functions of luminosity (LFs) enables SFRD measurement. However, due to noise and sample selection, these number densities are plagued by incompleteness and bias. The main goal of this project is to estimate the full completeness of my sample, including data depth, slit selection, and emission line recovery. To correct the incompleteness from my signal noise, I create a spectral simulation grid with *Pandeia*, calculating the percentage of objects my pipeline was unable to fit. To correct the incompleteness from my sample selection, I cross-match my sample with photometric catalogs covering the same fields to calculate the percentage of objects that do not have observed spectra. These data corrections create a more accurate lower limit to LFs up to z $\approx$ 7. Finally, I discuss expansions of this work that could refine these results by accounting for the effects of dust in my sample and further constraining the high-redshift SFRD.

1 Introduction

Modeling the history of the Universe through galactic evolution is a major field of study within modern astronomy (Sobral et al., 2015; Khostovan et al., 2020; Nagaraj et al., 2023; Lin et al., 2025). The Epoch of Reionization (EoR) describes a period of time $\sim$1Gyr after the Big Bang, during which high-energy radiation reionized the intergalactic medium of inert hydrogen gas that dominated the Universe during its preceding Dark Ages. Much of this high-energy radiation was emitted by galaxies forming new stars. Calculating the star formation rate density (SFRD) of these early galaxies is important, as it tracks the overall growth of galaxies over cosmic time (Sobral et al., 2015; Khostovan et al., 2020; Nagaraj et al., 2023).

Galaxies that emit the H-α emission line are strong tracers of star formation, so by analyzing the emission spectra of these galaxies, I can calculate SFRD (Sobral et al., 2015; Khostovan et al., 2020; Nagaraj et al., 2023; Lin et al., 2025). This is well-established, and is due to ionizing radiation emitted by young, massive stars triggering hydrogen recombination (Lin et al., 2025). We observe this effect by virtue of the H-α emission line.

The spectroscopic data I use is from the Red Unknowns: Bright Infrared Extragalactic Survey (RUBIES), and covers galaxies in the Extended Groth Strip (EGS) and Ultra Deep Survey (UDS) fields in the sky (de Graaff et al., 2025; Wang et al., 2024; Collaboration et al., 2022). RUBIES contains spectral data from the James Webb Space Telescope's Near-Infrared Spectrograph (JWST NIRSpec), which allows for the observation of spectra with observed wavelengths of $\sim$1-5 microns (JWS).

By analyzing a sample of 504 H-α detected galaxies at redshifts $2 < z < 7$ from RUBIES, I can calculate SFRD. After measuring the H-α luminosity of each galaxy in my sample using the Bayesian fitting pipeline *emcee*, I calculate the number density of galaxies at different luminosities and redshifts (Foreman-Mackey et al., 2013). However, my data requires completeness corrections to account for the signal-to-noise ratio (SNR) cutoff in my *emcee* pipeline, and for the sample selection bias of NIRSpec. These are referred to as line detection completeness corrections and observational completeness corrections respectively, and are calculated using a spectral simulation constructed in *Pandeia* and catalog cross-matching conducted with AstroPy (Pontoppidan et al., 2016; Finkelstein et al., 2023; JWS). After correcting the data, I can more accurately integrate the luminosity functions (LFs) to calculate SFRD, building a more constrained lower limit on the impact star formation had on the growth of galaxies over early cosmic time.

2 Sample Selection and Fitting

The JWST NIRSpec instrument includes a dispersive prism tool through which it observes the emission spectra of galaxies from observed wavelengths between 0.6-5.3 microns (JWS). RUBIES contains 1,951 of these spectroscopic objects from which I create my sample (de Graaff et al., 2025; Wang et al., 2024). To analyze H-α detected galaxies specifically, I select only those spectra that can include H-α between observed wavelengths of 0.6-5.3 microns. I calculate the redshift of a galaxy based on the known rest-frame wavelengths of several emission lines and Equation 1. Here, z is the redshift, λ_{obs} is the observed wavelength, and λ_{rest} is the rest-frame wavelength. After cutting out objects outside the redshift range $2 < z < 7$, my sample consists of 990 H-α detected RUBIES galaxies.

$$z = \frac{\lambda_{\text{obs}} - \lambda_{\text{rest}}}{\lambda_{\text{rest}}} \tag{1}$$

Each of these galaxies contains H-α within the observed wavelength range of 0.6-5.3 microns. Thus, I fit each emission line to a Gaussian curve using a Markov Chain Monte Carlo algorithm in Python called *emcee* (Foreman-Mackey et al., 2013). The *emcee* algorithm reads in the initial fit parameters defining a Gaussian curve (center, width, and amplitude), and uses those as starting points for walkers. Each walker explores a parameter space constrained by preset priors, and calculates the likelihood of each parameter jump being an appropriate fit to the data. After optimizing the number of walkers and parameter jumps, my algorithm returns the final parameter fits for each walker. By taking the median parameter fits from these walkers, I fit Gaussians to the 990 H-α emission lines. One such fit is shown in Figure 1.

I create an SNR cutoff to make sure that the Gaussian fits to each emission line are accurate, and are not being skewed due to large amounts of noise. If the SNR is less than 5, I mark the *emcee* parameter fits as not trustworthy enough, and I remove these galaxies from my sample. This leaves me with 504 H-α detected galaxies for which I have Gaussian fits for in flux units of Janskys (Jy; see Appendix B), which I convert into units of ergs/second/cm^2. By integrating

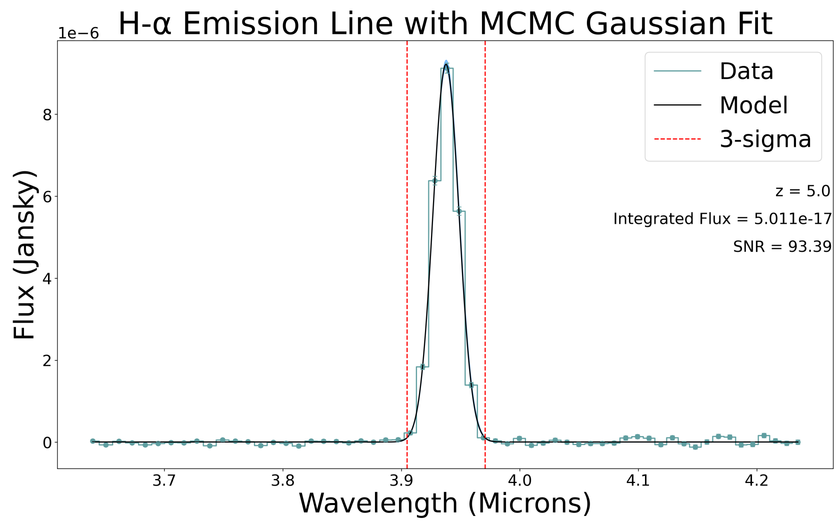

Figure 1: An example H-α emission line fit using the *emcee* Gaussian fitting pipeline (Foreman-Mackey et al., 2013). This particular galaxy has a redshift of 5.0. The peak flux point on this emission line has an SNR over 93, meaning that it is a very bright line and is easily fit, allowing for accurate calculation of its integrated flux.

under each Gaussian curve, I calculate the integrated flux of H-α in each galaxy. Using Equation 2, I calculate the luminosity for each source in units of ergs/second, and split these sources into 5 redshift bins centered at 2.5, 3.5, 4.5, 5.5, and 6.5. Here, L is the luminosity in ergs/second, F is the integrated flux, and D_L is the luminosity distance, which is calculated with an integral from the Astropy *Cosmology* library (Collaboration et al., 2022). Further information can be found in Appendix Section B.

$$L = 4\pi D_L^2 F \tag{2}$$

From here, I take each redshift bin of sources and calculate the number of objects that emitted H-α at different luminosities. Figure 2 shows the number of galaxies per unit volume at different luminosity values for each redshift bin. The volume calculations are conducted using Equation 3, with the size of the inclusive volume sphere varying with redshift. Here, V_c is the comoving volume, and D_c is the comoving distance, calculated by an integral from the Astropy *Cosmology* library (Collaboration et al., 2022). Further information can be found in Appendix Section B.

$$V_c(z) = \frac{4\pi}{3} D_C^3(z) \tag{3}$$

The plots displayed in Figure 2 are my LFs, and can be fit and integrated to calculate SFRDs. However, these LFs are lower limits and are not very accurate before data corrections, as they are plagued by incompleteness. Incompleteness here refers to my sample not being fully representative of the H-α detected galaxies that exist in the EGS and UDS fields (Taylor et al., 2024). It is important to characterize how many sources are and are not detected due to limitations in my data and methodology, as the ultimate goal is to determine an overall number density of sources. In this project, I address this incompleteness in two ways: line detection completeness corrections and observational completeness corrections (Taylor et al., 2024). These two data corrections are discussed in Sections 3 and 4.

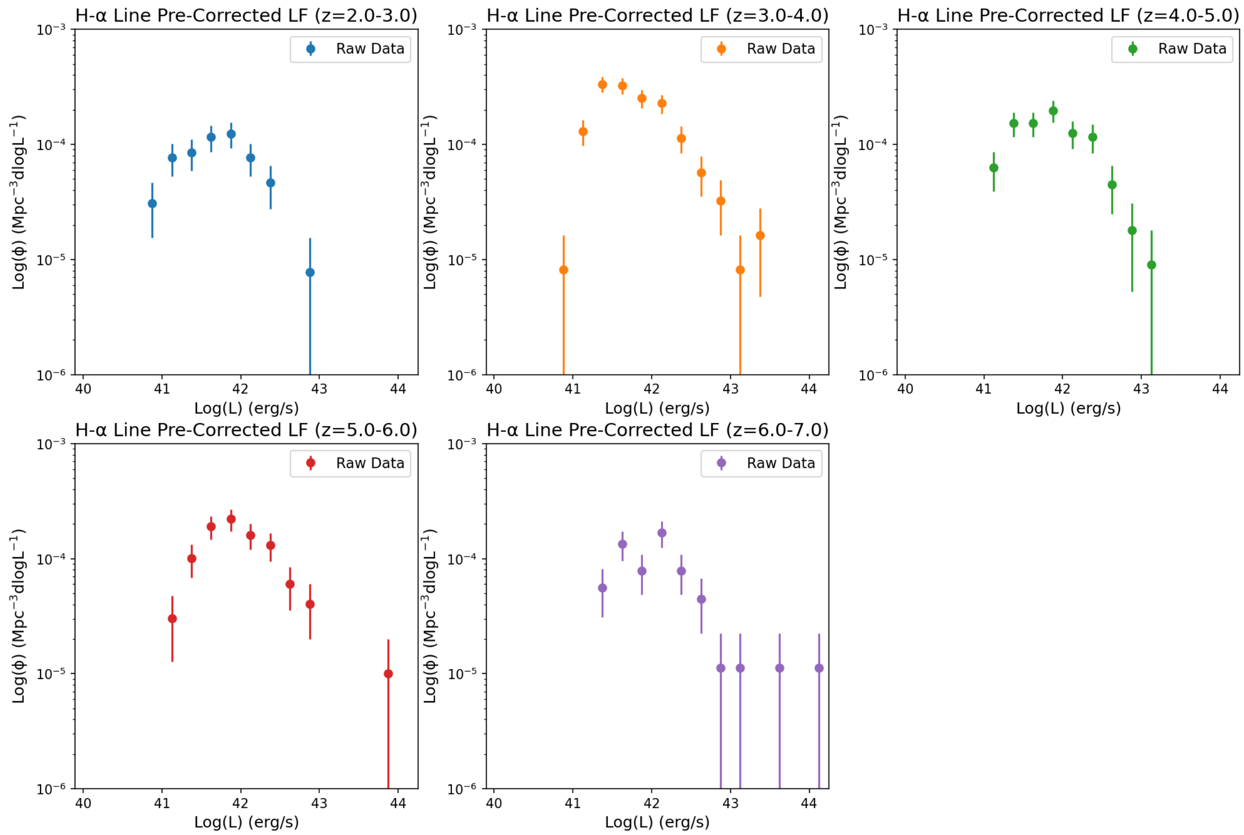

Figure 2: Pre-corrected LFs are shown for 5 different redshift ranges, and serve as a lower-limit constraint for my final LFs in Section 5. These display the number of galaxies per unit volume in each luminosity bin, with error bars calculated using a square root uncertainty per unit volume.

3 Line Detection Completeness Corrections

Line detection incompleteness is the inability to robustly detect H-α in every galaxy in the EGS and UDS fields. This is due to the *emcee* fitting algorithm not returning reasonable parameters for every H-α emission spectrum processed through it, as defined by the SNR>5 cutoff established earlier. If the noise in the emission spectra is too high to fit H-α (ie. the SNR<5), that galaxy is not admitted into my sample and contributes to the line detection incompleteness. This means that not every spectrum with H-α in its redshift range is included in my sample, leading to incompleteness. The line detection completeness percentage varies across redshifts and luminosities, so calculating and correcting for it is not a straightforward process.

The largest challenge is due to not having a control sample of galaxies that cover my entire redshift and luminosity range evenly (which would result in an inherently complete sample). To get around this, I create a simulation that covers all redshift and luminosity bins of interest using the JWST's *Pandeia* software package in Python (Taylor et al., 2024; Pontoppidan et al., 2016).

The *Pandeia* engine allows for JWST spectra to be simulated across a range of redshifts and luminosities to conduct line detection completeness corrections. User documentation is publicly available on the JWST website (ETC; Pontoppidan et al., 2016). My particular use of *Pandeia* is fleshed out as follows.

The JWST has many different instruments through which it collects data, both spectroscopically and photometrically. Furthermore, instruments possess many different operating modes and configurations across different observed wavelength ranges and resolutions. After building and specifying the proper Python environment to run *Pandeia*, I apply its default build functions to my pipeline. These specify that my spectral simulations are to be done using the "jwst" telescope, the "nirspec" instrument, and the "multi-object spectroscopy (mos)" filter (JWS; Pontoppidan et al., 2016).

After selecting the default build with my instrumental specifications, I run *Pandeia*'s "perform_calculation" function to read out the default parameter specifications that accompany my instrumental specifications. These include the normalization type, instrument type, filter, and disperser settings. From the default values, I edit several parameter specifications to match those of my sample of 504 H-α detected galaxies. I set the disperser to "prism" and remove the "mos" filter, which both match the JWST NIRSpec parameters of my sample and cover the observed wavelength range of 0.6-5.3 microns (JWS; Pontoppidan et al., 2016; Wang et al., 2024; de Graaff et al., 2025).

At this point, *Pandeia* can read in wavelength and flux arrays, and create a curve that would represent a spectrum at any given redshift value. However, the units of this curve are in "Counts", the number of photons detected per second per pixel (Pontoppidan et al., 2016). While I am still able to simulate spectra that appear visually accurate, these need to be in luminosity units of ergs/s to be used for data corrections. Thus, I need to create a calibration curve that can apply to each of the simulated spectra in order to convert the output units (Taylor et al., 2024).

To create this calibration curve, I feed in a wavelength array spanning the range of 0.5-6 microns, (slightly wider than each real spectrum in my sample), and a flux array of 1 Jansky held flat across the wavelength array. This allows the calibration curve to convert flux units while maintaining the shape a spectra captured with my instrumental specifications would take (Pontoppidan et al., 2016). This calibration curve is shown in Figure 3, with units of Counts/Jansky, so that dividing an output in units of Counts by the calibration curve results in the proper flux units of Janskys.

With the calibration complete, I move on to simulating spectra. I first establish a redshift range of 2<z<7 to match my sample, evenly spacing 11 redshift bins. I also establish an integrated flux range based on the maximum and minimum integrated flux values calculated from the *emcee* Gaussian fit curves described in Section 2, evenly spacing out 31 integrated flux bins. Finally, I establish an initial full-width-half-maximum (FWHM) value of 10nm. This FWHM describes the width of the simulated H-α emission line at half its amplitude (Foreman-Mackey et al., 2013). These values are used to calculate the Gaussian parameters for a simulated H-α emission line. For a given redshift, integrated flux, and FWHM value, the center, spread, and amplitude of a simulated Gaussian curve can be calculated using Equations 4, 5, and 6, respectively. Here, μ is the central wavelength of the Gaussian, λ_{H_α} is the rest-frame wavelength of the H-α emission line (6562.8 nm), σ is the spread of the Gaussian, w is the FWHM of the Gaussian, A is the amplitude of the Gaussian, and other variables remain consistent from previous equations. In essence, I construct a model H-α emission line based on tunable input parameters rather than by fitting real data.

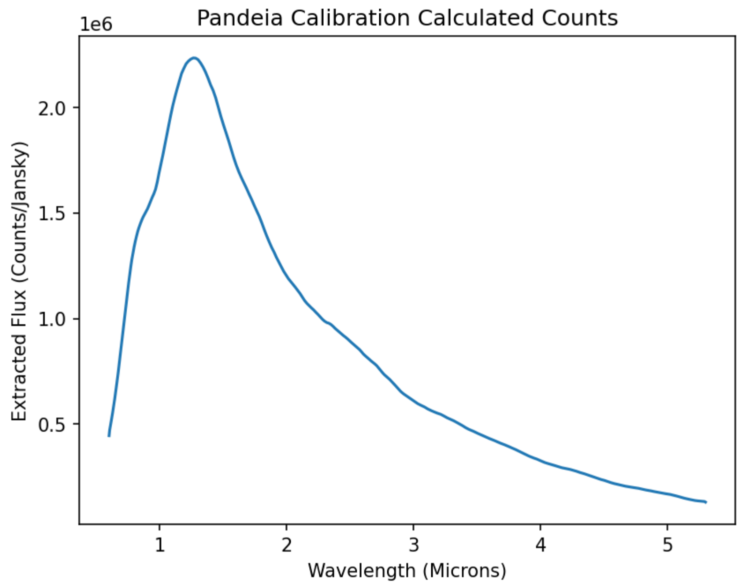

Figure 3: The calibration curve from *Pandeia* in units of Counts/Jansky that I divide from each simulated spectra. The shape of the curve is indicative of the shape a spectrum captured with my instrumental specifications would take (JWS; Taylor et al., 2024).

$$\mu = \lambda_{H_\alpha}(1+z) \tag{4}$$

$$\sigma = (1+z)\left(\frac{w}{2\sqrt{2\ln 2}}\right) \tag{5}$$

$$A = \frac{F}{\sigma\sqrt{2\pi}} \tag{6}$$

By iterating through the redshift and integrated flux ranges, I create a grid of model H-α lines with known Gaussian centers, amplitudes, and widths. I run this output through *Pandeia*'s "perform_calculation" function, which manipulates the Gaussian using the instrumental specifications established earlier to mimic data collected through NIRSpec's PRISM disperser. This produces a wavelength, flux, and flux error array for each spectrum I applied the function to. I divide the output flux values of these simulated spectra by the calibration curve to convert to proper units (JWS; Pontoppidan et al., 2016). The next step to make these simulations realistic is to simulate noise similar to that of the real spectra in my sample(de Graaff et al., 2025; Wang et al., 2024).

To simulate noise, I select from a normal random distribution between 0 and the flux error values for each point along the simulated flux error arrays with NumPy. I add each of these selected flux error values to the corresponding flux values for every point in the simulated flux arrays, generating statistically realistic noise. Pandeia sets the random seed to the same value for each noise simulation. Since I need to run several noise iterations for each redshift and integrated flux value in the grid space, I reset the random seed for each iteration to ensure that the noise patterns did not all follow the same structure. This ensures true random noise iterations within the range of each point's flux error.

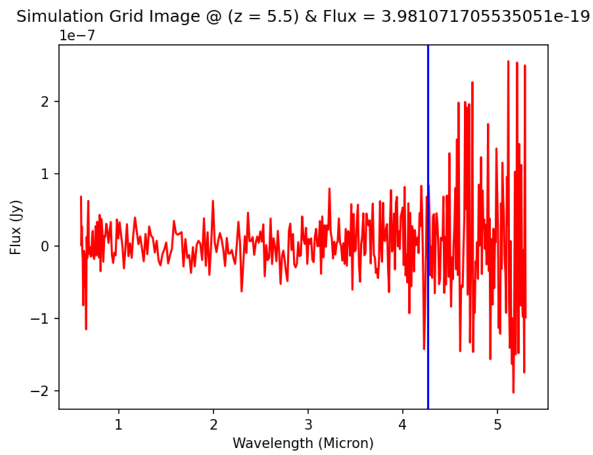

Figure 4: An example of a *Pandeia* simulated spectrum with a random noise iteration applied. The integrated flux input for this spectrum is quite low, and the vertical line marking where the H-α emission line should be shows an extremely low SNR, where the noise masks the signal.

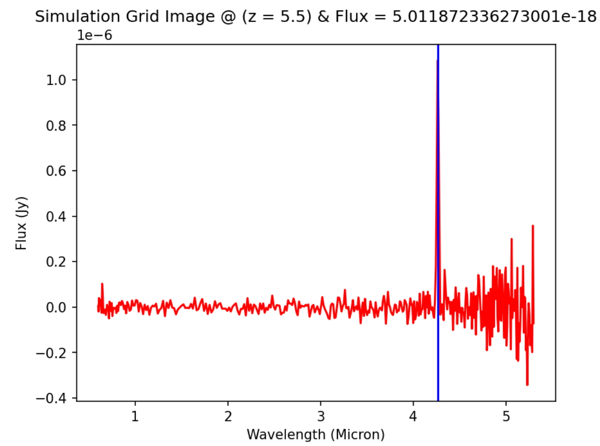

Figure 5: A second example of a *Pandeia* simulated spectrum with a random noise iteration applied. The integrated flux input for this spectrum is quite high, and the vertical line marking where the H-α emission line should be shows a relatively high SNR, where the signal is clear and easy to fit. Figures 4 and 5 both show spectra at a redshift of z=5.5.

At this point, I have a fully functioning framework for creating mock observations for any number of redshift and integrated flux combinations. I can now take the mock observation grid space of different redshift and integrated flux inputs, calculate Gaussian parameters for each grid element, and create an output of wavelength, flux, and flux error arrays with several noise iterations for each element. Figures 4 and 5 display two of these *Pandeia* simulated spectra at the same redshift. Figure 4 displays a spectrum with a low H-α flux, while Figure 5 displays one with a high H-α flux, both within the specified ranges of my sample (de Graaff et al., 2025; Wang et al., 2024).

At this point, I begin to calculate the line detection completeness percentage of the simulated grid space. With 11 redshift bins and 31 integrated flux bins, I have 341 different bin combinations for which I calculate the line detection completeness. For the initial stage of testing, I create 5 mock H-α lines, each with a unique resampling of simulated noise, for each of the 341 grid elements, essentially creating 5 different spectra for each redshift and integrated flux value (Taylor et al., 2024). I run each of these spectra through the same *emcee* pipeline laid out in Section 2, and calculate the SNR of the fitted Gaussian for each one. Each noise iteration with a SNR>5 is marked as successfully detected. Each element's completeness percentage is defined as the percentage of its noise iterations that are successfully detected.

Several elements have 0% completeness, while several have 100% completeness, corresponding with the lowest and highest integrated flux values, respectively. This is expected, as bright objects inherently have more signal in their data, while the faintest objects lack sufficient signal to

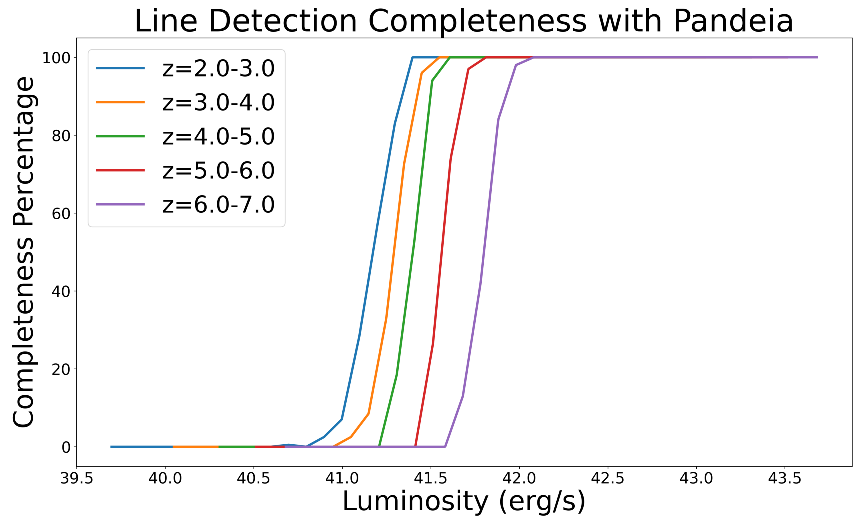

Figure 6: The line detection completeness percentage is shown as an interpolated function of luminosity across different redshift bins. There is a clear cutoff where my *emcee* pipeline fits essentially none of the spectra, and another were it fits essentially all of them (Foreman-Mackey et al., 2013; Pontoppidan et al., 2016). The shape of each function is similar, but the cutoffs occur at brighter values for higher redshifts.

be robustly detected (JWS; de Graaff et al., 2025; Wang et al., 2024). For the elements in between, I run a second stage of testing, simulating between 20 and 100 noise iterations for each element through the *emcee* pipeline to improve the precision of the line detection completeness percentages (Taylor et al., 2024). After running all the simulated spectra with the necessary noise iterations through my fitting pipeline, I compute completion percentages for the detection of H-α across the redshift and integrated flux ranges of my entire sample. Figure 6 displays these completeness percentages, which are applied to my sample in Section 5.

4 Observational Completeness Corrections

Observational incompleteness is the phenomenon of NIRSpec not observing spectra for every galaxy in the EGS and UDS fields. This occurs due to the physical limitations of the NIRSpec instrument, as it can only observe a fraction of the objects in its field of view in a single observation (JWS). While NIRSpec has thousands of shutters, we cannot open them all, as this would cause spectra from different objects to all overlap (JWS). Thus, the 1,951 objects within the RUBIES spectral sample represent only a fraction of the 131,720 sources observed photometrically with the JWST NIRCam instrument(de Graaff et al., 2025; Wang et al., 2024). (Photometric observation refers simply to the imaging of a galaxy in these fields). These sources have been tabulated into a catalog called UNICORN, and will be referred to as such from here on out (Finkelstein et al., 2023).

To correct for this incompleteness, I calculate the percentage of UNICORN objects at different redshift and magnitude bins that I also observe in the RUBIES catalog. To create these correction bins, I first calculate the photometric magnitude of each object, which in astronomy is defined as

Redshift Range	Flux Filter
$0 < z < 0.54$	F090W
$0.55 < z < 0.98$	F115W
$0.99 < z < 1.60$	F150W
$1.61 < z < 2.52$	F200W
$2.53 < z < 3.72$	F277W
$3.73 < z < 4.96$	F356W
$4.97 < z < 7.00$	F444W

Table 1: Redshift ranges and corresponding NIRSpec flux filters

the object's apparent brightness in the sky. Magnitude can be calculated from flux using Equation 7. Here, m is the magnitude. However, the flux that best defines an object's magnitude is measured through different filters depending on the object's redshift, as we want to sample the brightness of each spectrum in a wavelength range that covers the H-α line (JWS). Table 1 shows the optimal filter for which I extract the flux of each UNICORN object (Finkelstein et al., 2023).

$$m = -2.5 \log_{10}(F) + 31.4 \tag{7}$$

After calculating the magnitude of each object, I separate all the UNICORN objects into 5 redshift bins (as done in Sections 2 and 3) and 24 magnitude bins ranging from 16-40. Then, I use the AstroPy package SkyCoord to match objects in the UNICORN catalog to the RUBIES catalog based on their celestial coordinates: right ascension (RA) and declination (DEC) (Collaboration et al., 2022). If the RA and DEC values from both catalogs are within a 0.25 arcseconds of each other, I mark these objects as being the same(Collaboration et al., 2022). Once I have the number of matches, I divide this by the total number of UNICORN objects in each redshift+magnitude bin to calculate the fraction of objects in the UNICORN catalog that were observed by NIRSpec. This observational completeness percentage is shown in Figure 9 as a function of magnitude. Once I have the observational completeness percentage as a function of redshift and magnitude, I calculate a "weight" for each source in my sample of 504 RUBIES H-α detected galaxies. Based on the magnitude of each object in my sample, calculated again with the appropriate flux from Table 1 using Equation 7, I assign a weight to each one based on the completeness percentage of its redshift+magnitude bin in Figure 7. For example, an object in a bin with 50% observational completeness has a weight of 2, meaning that there are 2 objects present for the one that I have a spectrum of in my sample.

5 Results

Now, I apply the line detection and observational completeness corrections from Sections 3 and 4 to Figure 2 to correct the LFs, allowing for more accurate fitting and SFRD calculations. First, the weights calculated in Section 4 are applied to each source in the LFs. This accounts for NIRSpec only containing spectra for a fraction of galaxies in the EGS and UDS fields, causing the number densities of galaxies on the y-axis to increase accordingly (Taylor et al., 2024). Then, I divide each redshift+luminosity bin by the corresponding line detection completeness percentage from

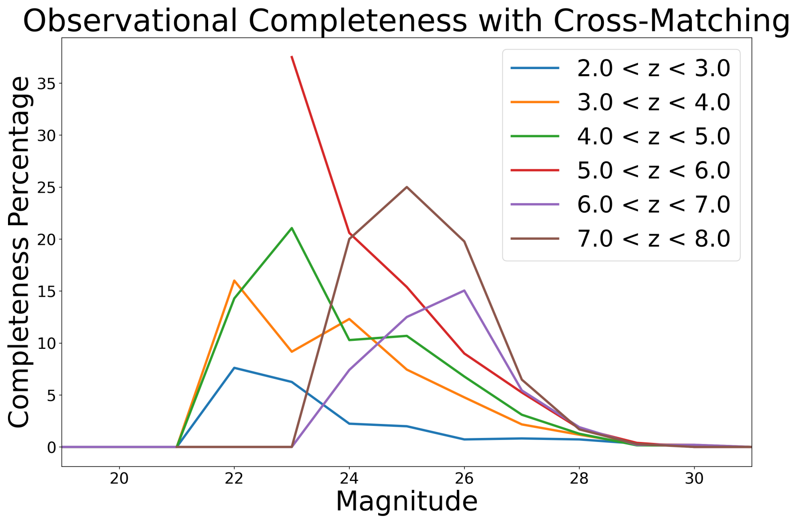

Figure 7: The observational completeness percentage is shown as a function of magnitude across different redshift bins (JWS; Finkelstein et al., 2023). In astronomy, a smaller magnitude value denotes an object appearing brighter in the sky. This completeness correction shows a similar pattern across redshifts, where NIRSpec only observed spectra for objects in my sample between magnitudes of 21 and 29, with varying completeness (de Graaff et al., 2025; Wang et al., 2024). These corrections are applied with weights on a source-by-source basis, where weights are dependent on the redshift+magnitude bin of a source.

Section 3. This accounts for my *emcee* fitting algorithm being unable to fit H-α lines with too much noise, causing each bin to increase accordingly (Taylor et al., 2024; Lin et al., 2025). Thus, the observational completeness corrections are applied on a source-by-source basis, while the line detection completeness corrections are applied on a bin-by-bin basis. Figure 8 shows the final results of these data corrections with the original points from Figure 2. There is a clear increase in the calculated number densities of galaxies across my entire sample, ranging from 0 to 3 orders of magnitude.

Figure 6 displays the limitations of my *emcee* fitting algorithm discussed in Section 2, while Figure 7 displays the incompleteness of my sample due to the instrumental limitations of the JWST's NIRSpec instrument (Foreman-Mackey et al., 2013; JWS). Figure 8 ties these data corrections together, creating more accurate LFs to analyze, and includes best-fit curves further discussed in Section 6. To calculate SFRD, these LFs need to be fit and integrated (Lin et al., 2025; Sobral et al., 2015; Khostovan et al., 2020; Nagaraj et al., 2023). Therefore, these data corrections have increased the calculated number densities of galaxies across the luminosity and redshift ranges of my sample to better represent the true populations, both detected and undetected, and in turn have increased the calculated SFRD.

6 Discussion

The application of these corrected LFs is the calculation of SFRD across different redshifts. To do so, I integrate under the curve defining the LFs in each redshift bin (Kennicutt, 1998). However,

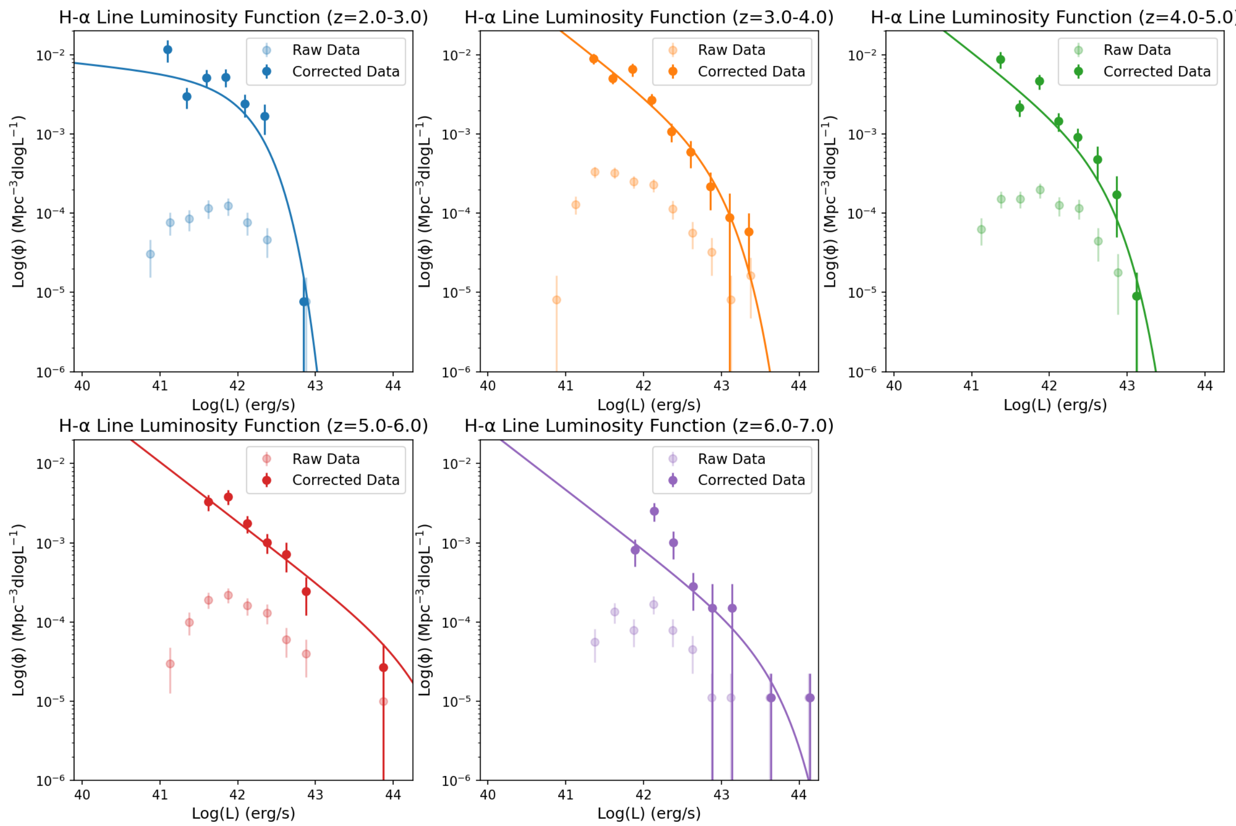

Figure 8: Post-corrected LFs are shown for 5 different redshift ranges, which apply the line detection and observational completeness corrections from Sections 3 and 4. The dim points in the background are the pre-corrected LF points from Figure 2, while the bright points are corrected, and are fit to Schechter curves of best-fit (Lin et al., 2025; Sobral et al., 2015; Khostovan et al., 2020; Nagaraj et al., 2023). The 2 highest redshift bins have a much later "knee" than the lower 3 redshift bins, likely due to contamination in the sample or the fit constraints being too loose.

I must first fit the corrected LFs to a curve called a Schechter Function, a combination of a faint-end power law and a bright-end exponential decay law, which is shown in Equation 8 (Lin et al., 2025; Sobral et al., 2015; Khostovan et al., 2020; Nagaraj et al., 2023). Here, $\phi(L)$ is the number density of galaxies as a function of luminosity, ϕ^* is the characteristic flux, L^* is the characteristic luminosity, and α is the faint-end slope. "Characteristic" here simply means the point at which the bright-end law begins to dominate the faint-end law. I use the same *emcee* fitting pipeline as discussed in Section 2, but replace the Gaussian fitting curve with the Schechter function fit. Thus, the median output parameter fits serve as the Schechter parameters used to fit the corrected LFs in Figure 8 (Foreman-Mackey et al., 2013). Figure 9 shows the fitted Schechter curves for each LF together. Then, I integrate each of these LFs between 41.25 and 45.0 ergs/s on the x-axis.

$$\phi(L)\,dL = \phi^* \left(\frac{L}{L^*}\right)^{\alpha} \exp\left(-\frac{L}{L^*}\right) \frac{dL}{L^*} \tag{8}$$

Using Equation 9, I convert the LF integrals into units of SFRD (Kennicutt, 1998). Here, the

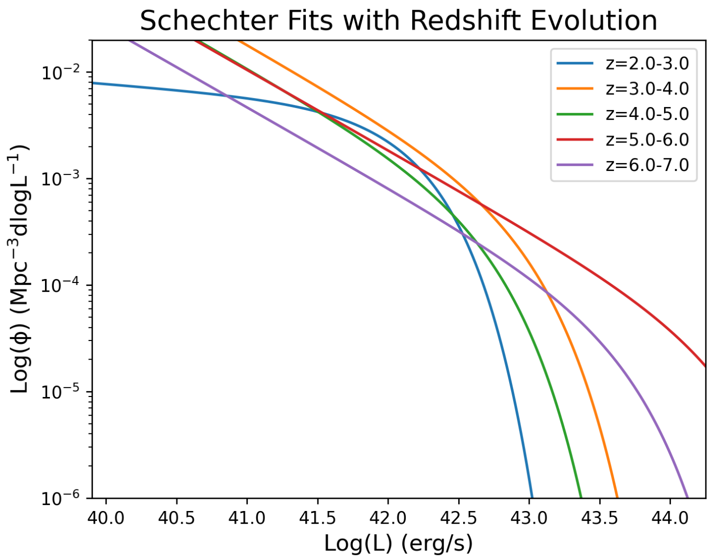

Figure 9: The Schechter fits from Figure 8 are shown on a single plot to emphasize the redshift evolution of the post-corrected LFs.

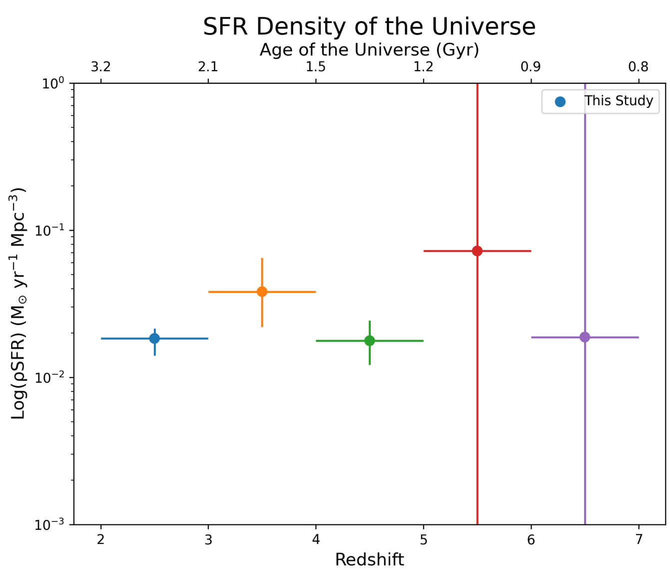

Figure 10: The SFRD is calculated by integrating each of the Schechter fits from Figure 9. The top axis shows the age of the Universe as redshift varies. There is not a clear increase or decrease in the SFRD across redshifts, but the error bars for the high-redshift bins are quite high, and need to be better constrained in the future.

SFR is given in units of solar masses per year, and L_{H_α} is the luminosity integral under the Schechter curve (Sobral et al., 2015; Lin et al., 2025; Khostovan et al., 2020; Nagaraj et al., 2023). The SFRD values for each redshift bin are shown in Figure 10. The error bars are calculated from integrating all Schechter curves from the entire *emcee* parameter space, and then keeping the fits within one standard deviation of the median fit as reasonable uncertainty. These SFRD values are higher and more accurate due to the data corrections undertaken in this project.

$$\text{SFR}\ (M_\odot\,\text{yr}^{-1}) = 7.9 \times 10^{-42}\ L_{\text{H}\alpha}\ (\text{erg}\,\text{s}^{-1}) \tag{9}$$

7 Summary

1. Measuring the SFRD of the early Universe helps astronomers to model the growth of galaxies over cosmic time. The H-α emission line is a strong tracer of star formation, so by measuring its flux I can calculate SFRD. My sample consists of 504 H-α detected galaxies in the RUBIES catalog, which contains NIRSpec spectra from the EGS and UDS fields.

2. I calculate the integrated flux of H-α using an *emcee* Gaussian fitting algorithm. After converting these fluxes into luminosities, I plot the number density of galaxies per unit volume at different luminosity bins, creating pre-corrected LFs at different redshifts.

3. Line detection incompleteness occurs when the H-α line's Gaussian fit has a SNR$>$5. To correct my data, I create a grid of simulated spectra across the redshift and luminosity ranges of my sample using *Pandeia*. I fit anywhere from 5-100 noise iterations per simulated grid element to a Gaussian using *emcee*, and calculated the percentage of objects per element that were successfully fit.

4. Observational incompleteness occurs because NIRSpec does not observe spectra for every galaxy in the EGS and UDS fields. I cross match the RUBIES catalog with a photometric catalog covering the same area called UNICORN, and calculate the percentage of objects at different magnitudes and redshifts that I have spectra of. I apply the corresponding correction weights to each object in my sample.

5. With corrected LFs, I fit the data to Schechter curves, and integrate under the fits to calculate SFRD between redshifts $2<z<7$. This allows us to have a lower-limit constraint on the effect early star formation had on the overall growth of galaxies over cosmic time. Further work on this project entails improving my sample by correcting for the effects of dust in NIRSpec spectroscopy, removing potential active black holes from my sample, and creating stricter priors to the *emcee* Schechter function fits.

8 Acknowledgments

I would like to thank Dr. Steven Finkelstein for serving as my faculty mentor on this research project, and aiding me in my research pursuits within his GEVIP research group. I would also like to thank Dr. Anthony Taylor for his insight on using *Pandeia* to simulate spectra, and Mr. Oscar Ortiz for his assistance in building the *emcee* fitting pipeline. I would finally like to thank the

RUBIES team for compiling their catalog of 1,951 PRISM objects with NIRSpec spectra from the JWST, and the rest of the JWST team for their work on the telescope's upkeep and data collection.

A Computer Code

The following public GitHub repository contains 4 different Jupyter notebooks involved with completing this research project: **https://github.com/rahul-shaji-v/star-formation-pandeia/tree/main**

This covers all of the scientific computation discussed throughout this paper. It also contains a poster creating a visualization of the broader application of this project, along with an animation displaying evolving simulated spectra from *Pandeia*.

B Further Equations

In Equation 3, the comoving volume V_c is calculated using the comoving distance D_c. This comoving distance is calculated using the Astropy *Cosmology* library, which calculates D_c using Equation 10. Here, Ω_m is the fraction of the Universe's mass-energy that is made up of matter (both baryonic and dark). A Ω_M less than 1 implies that the Universe is flat, and that 70% of the Universe's mass-energy is made up of dark energy. Additionally, H_0 is the Hubble constant, which is the increase in distance between two galaxies per unit of time due to the expansion of space in the Universe. Finally, c is the speed of light. The Astropy *Cosmology* library uses values of $\Omega_m = 0.3$, $c = 299,792.458$ km/s and $H_0 = 70\,\text{km s}^{-1}\text{Mpc}^{-1}$.

$$D_C(z) = c\int_0^z \frac{dz'}{H_0\sqrt{\Omega_m(1+z')^3 + (1-\Omega_m)}} \tag{10}$$

In Equation 2, the luminosity distance D_L is also calculated using the Astropy *Cosmology* library, which is shown in Equation 11. Here, D_c is the integral defined in Equation 10.

$$D_L(z) = (1+z)D_c(z) \tag{11}$$

In Section 2, the flux unit Jansky (Jy) is introduced. The common unit conversion is shown below in Equation 12.

$$1\,\text{Jy} = 10^{-23}\,\text{erg}\,\text{s}^{-1}\,\text{cm}^{-2}\,\text{Hz}^{-1} \tag{12}$$

The Hz^{-1} unit factor implies flux per unit frequency. I convert into flux per unit wavelength using Equation 13, where f_λ is the flux per unit wavelength, f_ν is the flux per unit frequency, and λ is the wavelength in angstroms.

$$f_\lambda = \frac{f_\nu\, c}{\lambda^2} \tag{13}$$

This converts flux into units of $\text{erg}\,\text{s}^{-1}\,\text{cm}^{-2}\text{Å}^{-1}$ Then, I integrate across the wavelength range to calculate integrated flux, with units of $\text{erg}\,\text{s}^{-1}\,\text{cm}^{-2}$.

This concludes the appendix covering a more detailed derivation of the luminosity distance, comoving distance, and astronomical unit conversions.

References

JWST ETC Pandeia Engine Tutorial user documentation. `https://jwst-docs.stsci.edu/jwst-exposure-time-calculator-overview/jwst-etc-pandeia-engine-tutorial#gsc.tab=0`. Accessed: 2025-04-19.

JWST Near Infrared Spectrograph user documentation. `https://jwst-docs.stsci.edu/jwst-near-infrared-spectrograph#gsc.tab=0`. Accessed: 2025-04-19.

The Astropy Collaboration, Adrian M. Price-Whelan, Pey Lian Lim, Nicholas Earl, Nathaniel Starkman, Larry Bradley, David L. Shupe, Aarya A. Patil, Lia Corrales, C. E. Brasseur, Maximilian Nöthe, Axel Donath, Erik Tollerud, Brett M. Morris, Adam Ginsburg, Eero Vaher, Benjamin A. Weaver, James Tocknell, William Jamieson, Marten H. van Kerkwijk, Thomas P. Robitaille, Bruce Merry, Matteo Bachetti, H. Moritz Günther, Thomas L. Aldcroft, Jaime A. Alvarado-Montes, Anne M. Archibald, Attila Bódi, Shreyas Bapat, Geert Barentsen, Juanjo Bazán, Manish Biswas, Médéric Boquien, D. J. Burke, Daria Cara, Mihai Cara, Kyle E Conroy, Simon Conseil, Matthew W. Craig, Robert M. Cross, Kelle L. Cruz, Francesco D'Eugenio, Nadia Dencheva, Hadrien A. R. Devillepoix, Jörg P. Dietrich, Arthur Davis Eigenbrot, Thomas Erben, Leonardo Ferreira, Daniel Foreman-Mackey, Ryan Fox, Nabil Freij, Suyog Garg, Robel Geda, Lauren Glattly, Yash Gondhalekar, Karl D. Gordon, David Grant, Perry Greenfield, Austen M. Groener, Steve Guest, Sebastian Gurovich, Rasmus Handberg, Akeem Hart, Zac Hatfield-Dodds, Derek Homeier, Griffin Hosseinzadeh, Tim Jenness, Craig K. Jones, Prajwel Joseph, J. Bryce Kalmbach, Emir Karamehmetoglu, Mikołaj Kałuszyński, Michael S. P. Kelley, Nicholas Kern, Wolfgang E. Kerzendorf, Eric W. Koch, Shankar Kulumani, Antony Lee, Chun Ly, Zhiyuan Ma, Conor MacBride, Jakob M. Maljaars, Demitri Muna, N. A. Murphy, Henrik Norman, Richard O'Steen, Kyle A. Oman, Camilla Pacifici, Sergio Pascual, J. Pascual-Granado, Rohit R. Patil, Gabriel I Perren, Timothy E. Pickering, Tanuj Rastogi, Benjamin R. Roulston, Daniel F Ryan, Eli S. Rykoff, Jose Sabater, Parikshit Sakurikar, Jesús Salgado, Aniket Sanghi, Nicholas Saunders, Volodymyr Savchenko, Ludwig Schwardt, Michael Seifert-Eckert, Albert Y. Shih, Anany Shrey Jain, Gyanendra Shukla, Jonathan Sick, Chris Simpson, Sudheesh Singanamalla, Leo P. Singer, Jaladh Singhal, Manodeep Sinha, Brigitta M. Sipőcz, Lee R. Spitler, David Stansby, Ole Streicher, Jani Šumak, John D. Swinbank, Dan S. Taranu, Nikita Tewary, Grant R. Tremblay, Miguel de Val-Borro, Samuel J. Van Kooten, Zlatan Vasović, Shresth Verma, José Vinícius de Miranda Cardoso, Peter K. G. Williams, Tom J. Wilson, Benjamin Winkel, W. M. Wood-Vasey, Rui Xue, Peter Yoachim, Chen Zhang, and Andrea Zonca. The astropy project: Sustaining and growing a community-oriented open-source project and the latest major release (v5.0) of the core package*. *The Astrophysical Journal*, 935(2):167, August 2022. ISSN 1538-4357. doi: 10.3847/1538-4357/ac7c74. URL `http://dx.doi.org/10.3847/1538-4357/ac7c74`.

Anna de Graaff, Gabriel Brammer, Andrea Weibel, Zach Lewis, Michael V. Maseda, Pascal A. Oesch, Rachel Bezanson, Leindert A. Boogaard, Nikko J. Cleri, Olivia R. Cooper, Rashmi Gottumukkala, Jenny E. Greene, Michaela Hirschmann, Raphael E. Hviding, Harley Katz, Ivo Labbé, Joel Leja, Jorryt Matthee, Ian McConachie, Tim B. Miller, Rohan P. Naidu, Sedona H. Price, Hans-Walter Rix, David J. Setton, Katherine A. Suess, Bingjie Wang, Katherine E. Whitaker, and Christina C. Williams. Rubies: a complete census of the bright and red distant universe with jwst/nirspec, 2025. URL https://arxiv.org/abs/2409.05948.

Steven L. Finkelstein, Gene C. K. Leung, Micaela B. Bagley, Mark Dickinson, Henry C. Ferguson, Casey Papovich, Hollis B. Akins, Pablo Arrabal Haro, Romeel Dave, Avishai Dekel, Jeyhan S. Kartaltepe, Dale D. Kocevski, Anton M. Koekemoer, Norbert Pirzkal, Rachel S. Somerville, L. Y. Aaron Yung, Ricardo Amorin, Bren E. Backhaus, Peter Behroozi, Laura Bisigello, Volker Bromm, Caitlin M. Casey, Oscar A. Chavez Ortiz, Yingjie Cheng, Katherine Chworowsky, Nikko J. Cleri, Michael C. Cooper, Kelcey Davis, Alexander de la Vega, David Elbaz, Maximilien Franco, Adriano Fontana, Seiji Fujimoto, Mauro Giavalisco, Norman A. Grogin, Benne W. Holwerda, Marc Huertas-Company, Michaela Hirschmann, Kartheik G. Iyer, Shardha Jogee, Intae Jung, Rebecca L. Larson, Ray A. Lucas, Bahram Mobasher, Alexa M. Morales, Caroline V. Morley, Sagnick Mukherjee, Pablo G. Perez-Gonzalez, Swara Ravindranath, Giulia Rodighiero, Melanie Rowland, Sandro Tacchella, Anthony J. Taylor, Jonathan R. Trump, and Stephen Wilkins. The complete ceers early universe galaxy sample: A surprisingly slow evolution of the space density of bright galaxies at z   8.5-14.5, 2023. URL https://arxiv.org/abs/2311.04279.

Daniel Foreman-Mackey, David W. Hogg, Dustin Lang, and Jonathan Goodman. emcee: The MCMC Hammer. , 125(925):306, March 2013. doi: 10.1086/670067.

Robert C. Kennicutt, Jr. Star Formation in Galaxies Along the Hubble Sequence. *ARA&A*, 36: 189–232, January 1998. doi: 10.1146/annurev.astro.36.1.189.

A. A. Khostovan, S. Malhotra, J. E. Rhoads, C. Jiang, J. Wang, I. Wold, Z. Y. Zheng, L. F. Barrientos, A. Coughlin, S. Harish, W. Hu, L. Infante, L. A. Perez, J. Pharo, F. Valdes, A. R. Walker, and H. Yang. A large, deep 3 $\deg^2$ survey of H α, [O III], and [O II] emitters from LAGER: constraining luminosity functions. , 493(3):3966–3984, April 2020. doi: 10.1093/mnras/staa175.

Xiaojing Lin, Eiichi Egami, Fengwu Sun, Haowen Zhang, Xiaohui Fan, Jakob M. Helton, Feige Wang, Andrew J. Bunker, Zheng Cai, Daniel J. Eisenstein, Daniel T. Jaffe, Zhiyuan Ji, Xiangyu Jin, Maria Anne Pudoka, Sandro Tacchella, Wei Leong Tee, Pierluigi Rinaldi, Brant Robertson, Yang Sun, Christopher N. A. Willmer, Chris Willott, Junyu Zhang, and Yongda Zhu. The luminosity function and clustering of hα emitting galaxies at $z \approx 4-6$ from a complete nircam grism redshift survey, 2025. URL https://arxiv.org/abs/2504.08028.

Gautam Nagaraj, Robin Ciardullo, William P. Bowman, Alex Lawson, and Caryl Gronwall. The hα and [o iii] λ5007 luminosity functions of 1.2 lt; z lt; 1.9 emission-line galaxies from hubble space telescope (hst) grism spectroscopy. *The Astrophysical Journal*, 943(1):5, January 2023. ISSN 1538-4357. doi: 10.3847/1538-4357/aca7bb. URL http://dx.doi.org/10.3847/1538-4357/aca7bb.

Klaus M. Pontoppidan, Timothy E. Pickering, Victoria G. Laidler, Karoline Gilbert, Christopher D. Sontag, Christine Slocum, Mark J. Sienkiewicz, Christopher Hanley, Nicholas M. Earl, Laurent Pueyo, Swara Ravindranath, Diane M. Karakla, Massimo Robberto, Alberto Noriega-Crespo, and Elizabeth A. Barker. Pandeia: a multi-mission exposure time calculator for JWST and WFIRST. In Alison B. Peck, Robert L. Seaman, and Chris R. Benn, editors, *Observatory Operations: Strategies, Processes, and Systems VI*, volume 9910 of *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series*, page 991016, July 2016. doi: 10.1117/12.2231768.

D. Sobral, J. Matthee, P. N. Best, I. Smail, A. A. Khostovan, B. Milvang-Jensen, J.-W. Kim, J. Stott, J. Calhau, H. Nayyeri, and B. Mobasher. Cf-hizels, an □10 deg2 emission-line survey with spectroscopic follow-up: Hα, [o iii] + hβ and [o ii] luminosity functions at z = 0.8, 1.4 and 2.2. *Monthly Notices of the Royal Astronomical Society*, 451(3):2303–2323, 06 2015. ISSN 0035-8711. doi: 10.1093/mnras/stv1076. URL `https://doi.org/10.1093/mnras/stv1076`.

Anthony J. Taylor, Steven L. Finkelstein, Dale D. Kocevski, Junehyoung Jeon, Volker Bromm, Ricardo O. Amorin, Pablo Arrabal Haro, Bren E. Backhaus, Micaela B. Bagley, Eduardo Bañados, Rachana Bhatawdekar, Madisyn Brooks, Antonello Calabro, Oscar A. Chavez Ortiz, Yingjie Cheng, Nikko J. Cleri, Justin W. Cole, Kelcey Davis, Mark Dickinson, Callum Donnan, James S. Dunlop, Richard S. Ellis, Vital Fernandez, Adriano Fontana, Seiji Fujimoto, Mauro Giavalisco, Andrea Grazian, Jingsong Guo, Nimish P. Hathi, Benne W. Holwerda, Michaela Hirschmann, Kohei Inayoshi, Jeyhan S. Kartaltepe, Yana Khusanova, Anton M. Koekemoer, Vasily Kokorev, Rebecca L. Larson, Gene C. K. Leung, Ray A. Lucas, Derek J. McLeod, Lorenzo Napolitano, Masafusa Onoue, Fabio Pacucci, Casey Papovich, Pablo G. Pérez-González, Nor Pirzkal, Rachel S. Somerville, Jonathan R. Trump, Stephen M. Wilkins, L. Y. Aaron Yung, and Haowen Zhang. Broad-line agn at $3.5 < z < 6$: The black hole mass function and a connection with little red dots, 2024. URL `https://arxiv.org/abs/2409.06772`.

Bingjie Wang, Joel Leja, Anna de Graaff, Gabriel B. Brammer, Andrea Weibel, Pieter van Dokkum, Josephine F. W. Baggen, Katherine A. Suess, Jenny E. Greene, Rachel Bezanson, Nikko J. Cleri, Michaela Hirschmann, Ivo Labbé, Jorryt Matthee, Ian McConachie, Rohan P. Naidu, Erica Nelson, Pascal A. Oesch, David J. Setton, and Christina C. Williams. RUBIES: Evolved Stellar Populations with Extended Formation Histories at z $\sim$ 7–8 in Candidate Massive Galaxies Identified with JWST/NIRSpec. , 969(1):L13, July 2024. doi: 10.3847/2041-8213/ad55f7.